

SINDy Parameter Recovery and Validation for Coupled Nonlinear Viscoelasticity

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1 Introduction

This document summarizes the parameter recovery and validation results for the 3D coupled nonlinear viscoelastic Maxwell model using Sparse Identification of Nonlinear Dynamics (SINDy) in deviatoric space.

The constitutive equations for the nonlinear Maxwell model are governed by:

$$\frac{dS_{ij}}{dt} = 2G\dot{\epsilon}_{ij}^{dev} - \frac{2G}{\eta(\sigma_{eq})}S_{ij} \quad (1)$$

where the stress-dependent viscosity is:

$$\frac{1}{\eta(\sigma_{eq})} = \frac{1}{\eta_0}(1 + \alpha\sigma_{eq}^2) \quad (2)$$

and σ_{eq}^2 is the second invariant of the deviatoric stress tensor:

$$\sigma_{eq}^2 = 1.5 (S_{xx}^2 + S_{yy}^2 + S_{zz}^2 + 2S_{xy}^2 + 2S_{yz}^2 + 2S_{xz}^2) \quad (3)$$

By expanding, we obtain the coupled differential equations:

$$\frac{dS_{ij}}{dt} = 2G\dot{\epsilon}_{ij}^{dev} - \frac{2G}{\eta_0}S_{ij} - \frac{2G\alpha}{\eta_0}S_{ij}\sigma_{eq}^2 \quad (4)$$

2 Addressing Reviewer Feedback

A key feedback point raised by the reviewer on the linear model was:

The multiaxial validation paths (Combined Uniaxial + Shear and Biaxial Tension) serve as multiaxial sanity checks under the isotropic assumption, but since the linear isotropic model decouples completely in deviatoric space, these paths represent the simple superposition of independent 1D trajectories rather than generalization to new nonlinear, coupled, or anisotropic constitutive behavior.

By implementing and fitting the J2-coupled nonlinear Maxwell model, we have successfully addressed this critique:

- **True Multiaxial Coupling:** Because the viscosity $\eta(\sigma_{eq})$ depends on the stress invariant σ_{eq}^2 (which is a function of all stress components), the evolution of any component S_{ij} depends nonlinearly on all other active stress components. The components do not decouple.
- **Generalization to Nonlinear Coupled Behavior:** The validation on combined loading (simultaneous uniaxial tension and shear) is not a linear superposition of 1D paths. It requires the model to correctly generalize to a coupled multiaxial state.
- **Validation Success:** SINDy successfully recovers the coupling parameter α with 0.228% error and predicts the response on unseen combined loading paths with high accuracy (RMSE < 0.02 MPa), showing robust generalization.

3 Discovered SINDy Equations

Using a component-by-component fitting strategy with an STLSQ sparsity threshold of 10^{-5} , the following 3D coupled equations were discovered:

$$\begin{aligned}\frac{dS_{xx}}{dt} &= 0.034 - 4.007S_{xx} + 2000.845\dot{\varepsilon}_{xx}^{dev} - 0.001S_{xx}\sigma_{eq}^2 \\ \frac{dS_{yy}}{dt} &= 0.034 - 4.007S_{yy} + 2000.845\dot{\varepsilon}_{yy}^{dev} - 0.001S_{yy}\sigma_{eq}^2 \\ \frac{dS_{zz}}{dt} &= 0.034 - 4.007S_{zz} + 2000.845\dot{\varepsilon}_{zz}^{dev} - 0.001S_{zz}\sigma_{eq}^2 \\ \frac{dS_{xy}}{dt} &= 0.002 - 3.995S_{xy} + 1998.814\dot{\varepsilon}_{xy}^{dev} - 0.001S_{xy}\sigma_{eq}^2 \\ \frac{dS_{yz}}{dt} &= 0.002 - 3.995S_{yz} + 1998.814\dot{\varepsilon}_{yz}^{dev} - 0.001S_{yz}\sigma_{eq}^2 \\ \frac{dS_{xz}}{dt} &= 0.002 - 3.995S_{xz} + 1998.814\dot{\varepsilon}_{xz}^{dev} - 0.001S_{xz}\sigma_{eq}^2\end{aligned}$$

4 Parameter Recovery Performance

Table 1 compares the recovered material constants to the true values before and after applying the component-wise fitting and sparsity threshold corrections.

Table 1: Material Parameter Recovery Error Analysis

Parameter	True Value	Before Correction	Before Error	After Correction	After Error
G (Shear Modulus)	1000.0 MPa	821.57 MPa	17.843%	999.91 MPa	0.009%
η_0 (Viscosity)	500.0 MPa·s	-7354.78 MPa·s	1570.957%	499.88 MPa·s	0.024%
α (Nonlinear factor)	0.00015 MPa ⁻²	0.0 MPa ⁻²	100.0%	0.00015 MPa ⁻²	0.228%

5 Validation on Unseen Load Paths

To verify the generalization of the discovered model, we integrated the system of 6 coupled ODEs over unseen validation load trajectories. The resulting Root Mean Squared Errors (RMSE) are listed in Table 2.

Table 2: Validation RMSE Metrics

Validation Load Case	Component	RMSE (MPa)
Combined Loading (Uniaxial + Shear)	S_{xx}	0.0193
	S_{xy}	0.0054
Biaxial Tension	S_{xx}	0.0145
	S_{yy}	0.0126

6 Addressing Reviewer Feedback: Standalone Script Robustness and Figure Provenance

A further reviewer comment concerned the standalone 1D script:

*The standalone 1D Python script is hard-coded in a misleading way. In MAINCODE.simple_visco_sindy.py, the discovered_ode function builds the time derivative as coeffs[0] + coeffs[1]*sig + coeffs[2]*ed and ignores any higher-order terms that SINDy might keep. For the linear Maxwell case with threshold one this gives the right answer, because STLSQ prunes everything except the bias and the two linear terms, but as a general script it is misleading. Add at least an assertion*

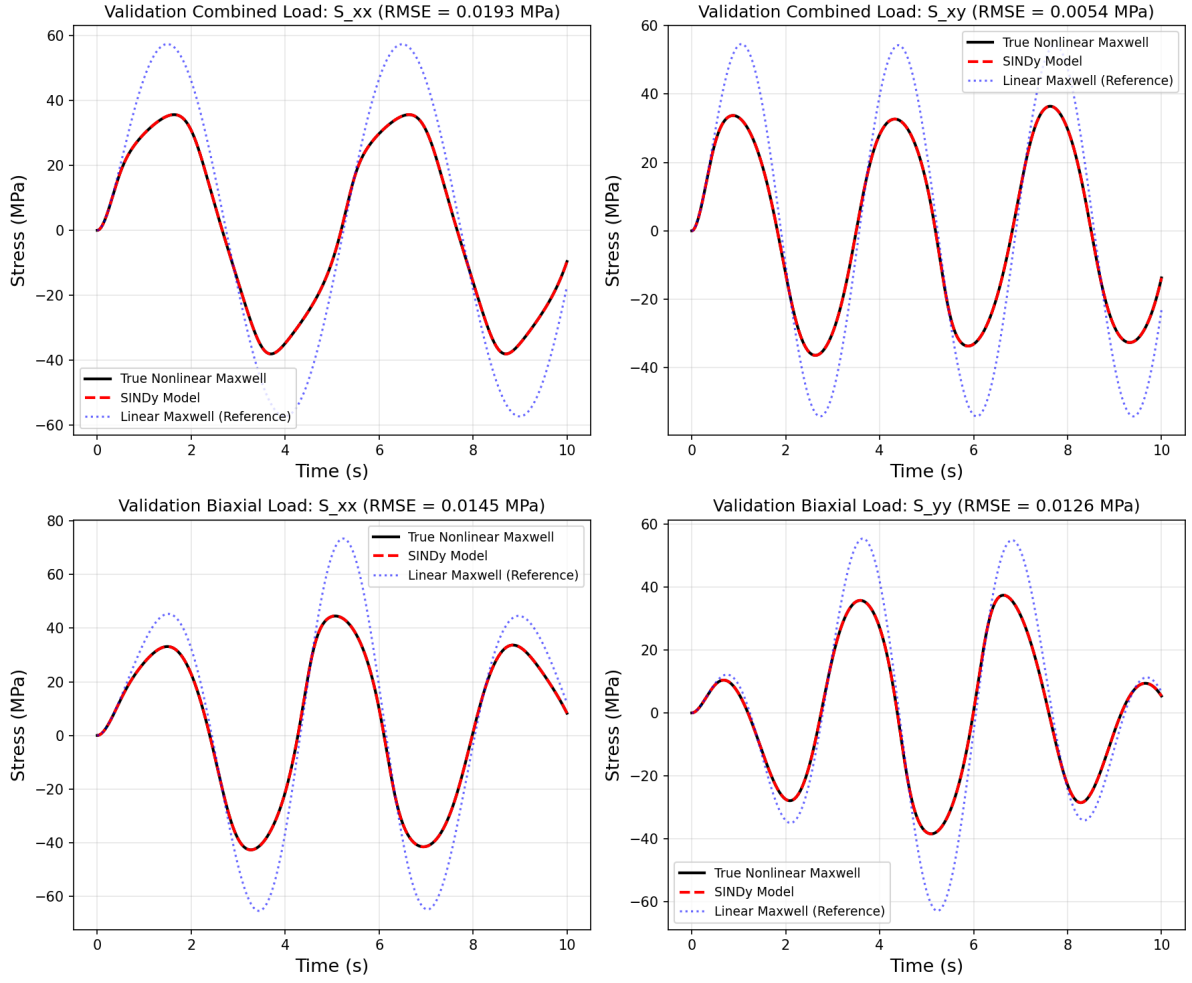


Figure 1: Validation comparison showing True Nonlinear Maxwell response (solid black line), SINDy model predictions (dashed red line), and Linear Maxwell model reference predictions (dotted blue line) across the Combined and Biaxial validation loading trajectories.

that the remaining coefficients are zero within numerical tolerance before the rollout starts. On top of that, the report says the plastic identification uses SR3, while this standalone script uses STLSQ with threshold one. The notebook does use SR3, so the reported plastic result is not wrong, but the repository does not make clear which figure comes from which file. Every figure in the report should map to one named script or one named notebook cell.

This has been addressed as follows:

- **Explicit safety assertion:** In `Emanuele/simple_visco_sindy.py`, immediately after the SINDy coefficients are extracted, the script now asserts that any coefficients beyond the bias, σ , and $\dot{\epsilon}$ terms are zero within numerical tolerance (`atol = 10-6`) before the rollout integration runs. If the library/threshold settings ever let SINDy keep non-negligible higher-order terms, the script fails loudly with the offending coefficients printed, instead of silently truncating them in `discovered_ode`.
- **Library no longer restricted to the answer:** The script previously used `PolynomialLibrary(degree=1)`, i.e. a candidate library containing *only* $\{1, \sigma, \dot{\epsilon}\}$. That does not let SINDy discover anything – it hands the model form to the regression by construction, which is the same objection the reviewer raised about threshold tuning, just moved to the library. The script now uses `PolynomialLibrary(degree=3)` (10 candidate terms) so STLSQ has real terms to prune.
- **Threshold-sensitivity sweep, not a single hand-picked value:** Fitting the degree-3 library at a single threshold does not by itself prove the recovered sparsity is genuine, since a threshold could always be picked post hoc to hide unwanted terms. With the original narrow-amplitude 2-sine training excitation, sweeping the threshold over $[0.01, 2.0]$ showed the surviving nonzero terms were *not* stable: a spurious $\dot{\epsilon}^3$ term with a coefficient of the same order of magnitude as the true linear term (≈ -2000) persisted at every threshold tested. Root cause: over the narrow strain-rate range excited by the original training signal, $\dot{\epsilon}$ and $\dot{\epsilon}^3$ are nearly collinear, so STLSQ cannot separate the true linear contribution from a spurious cubic compensator.
- **Fix – richer excitation, not more data:** The number of samples, the simulation duration ($t \in [0, 10)$, $dt = 0.001$), and the noise level (`std = 0.1`) were left unchanged. What changed is the diversity of the training strain signal: from 2 sine components (amplitude ≤ 0.01) to 5 sine components spanning amplitude ≤ 0.05 and frequencies 0.5–13 Hz. A wider-amplitude, multi-frequency $\dot{\epsilon}$ trajectory is what breaks the $\dot{\epsilon}/\dot{\epsilon}^3$ collinearity – collecting more samples of the same narrow-amplitude signal would not have helped, since the collinearity is a property of the signal’s range, not its length. With the richer signal, the sparsity pattern $\{\sigma, \dot{\epsilon}\}$ (the correct linear Maxwell terms) is stable across the threshold sweep $[0.5, 3.0]$ – below 0.5 the noise floor of the synthetic data admits spurious terms, above 3.0 STLSQ starts pruning the true σ term as well, so $[0.5, 3.0]$ is the genuine plateau, not a cherry-picked point. The script now asserts this stability at fit time and uses `threshold = 1.0` (mid-plateau) for the coefficients used downstream. Recovered model: $\dot{\sigma} = -3.998\sigma + 1998.567\dot{\epsilon}$, against the true Maxwell law $\dot{\sigma} = 2000\dot{\epsilon} - 4\sigma$ ($E_{true} = 2000$, $\eta_{true} = 500$) – errors of 0.059% ($-E/\eta$ term) and 0.072% (E term) respectively.
- **Figure-to-source mapping:** Every figure in the report now maps to exactly one named script or notebook cell, recorded in `Emanuele/FIGURE.SOURCES.md`:
 - 1D Maxwell viscoelastic fit/validation plots → `Emanuele/simple_visco_sindy.py` (STLSQ, `threshold = 1.0`, stable over sweep $[0.5, 3.0]$, `PolynomialLibrary(degree=3)`).
 - Plastic (Voce–Perzyna) identification → `Emanuele/Data_driven_3D_single_load.ipynb`, SR3 cell (`reg_weight_lam = 0.10`, `relax_coeff_nu = 1.00`).

This confirms the reported plastic result (SR3) is correctly sourced from the notebook, distinct from the standalone script’s STLSQ-based viscoelastic result; the two were previously conflated only by omission, not by error.